Eye as a Camera in XR: Reading Images, Colours and Eye Strain from a Consumer EEG Headband and a Consumer VR Headset

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***Abstract*—Mann showed in 2019 that the eye can be used as a camera: a square flickers at one position after another, the subject follows it, and the steady-state visual evoked potential (SSVEP) says how bright each position was. We move the method onto hardware a community health worker could carry: a Muse 2 headband with one cup electrode added over the visual cortex, and a Meta Quest 3S running the stimulus in its browser. We read a legible word in 18 minutes ($r = 0.91$ against time-shifted nulls of 0.23), eight colours as three calibrated colour planes (24 of 32 cells, reproduced over three sittings), and an acuity optotype drawn at random and sealed before the scan, named correctly from the EEG alone. The same recording gives an objective eye-strain reading: blinks stop while a cell flickers and burst when it goes dark, and the blink rate doubled between two sittings of one evening. We frame the result as a literacy-free visual screener in the tradition of low-cost mobile health tools, with a prompt to rest the eyes as its first intervention. All results are from one subject.**

***Index Terms*—SSVEP; EEG; extended reality; consumer electronics; brain-computer interface; mobile health; visual screening; eye strain; just-in-time adaptive intervention; wearable computing.**

I. Introduction

A camera needs a lens, a sensor and a readout. The eye has the first two; Mann's "eye as a camera" [1], [2] supplied the third. A square flickers at a fixed rate, the subject looks at it, and the visual cortex answers with an SSVEP whose strength follows the light the square puts into the fovea [3], [4]. Scan the square across an image and the sequence of strengths is the image. The 2019 paper read "NO CAMERAS" and a portrait with a laboratory amplifier and 85 minutes of scanning.

This paper asks what the method can do on hardware that costs less than a phone and needs no technician: a Muse 2 headband (about $250) with one cup electrode through its auxiliary input, a Meta Quest 3S (about $300) running the stimulus as a web page, and a laptop. Our interest comes from the mobile health tradition of low-cost sensing in the hands of community health workers [5]–[7]. A visual screen read from the EEG asks the subject to name no letter and press no button; they look at a dot. The same recording carries what the mobile health literature has long wanted from wearables, an objective reading of the subject's state [8], [9], here the eye strain that flicker produces.

We contribute: the first SSVEP imaging on a consumer VR headset with a consumer headband; colour by three calibrated planes; a sealed, objective acuity read-out; eye strain from the same recording; and the field engineering that made the method survive outside a laboratory, with a synthetic subject as regression test.

II. Background

The SSVEP is largest over the occipital midline (Oz) and strongest for stimuli in the central field [3], which is what makes a single flickering patch under the gaze one pixel per position [1]. SSVEP interfaces in head-mounted displays are established [10]; imaging on one, and on the Quest family, is not. The sweep VEP reads acuity without a verbal response [11]; our optotype read-out is a coarse, gaze-scanned relative of it. Chromatic VEPs differ by colour axis and temporal frequency [12], which is why our colour planes are calibrated separately. The Muse headband is validated for event-related potentials [13] but has no electrode near Oz; the added cup is what makes imaging possible (cup alone $r = 0.92$, ear electrodes alone 0.36). A VR headset over the Muse improves the contact rather than degrading it [14].

Fletcher's group has built a decade of diagnostics around sensors a phone already has, for workers who are not specialists: pulmonary screening from cough and lung sounds [7], [15], [16], child health tools [6], printed diagnostics read by the camera [17], thermal imaging with deep learning [18], [19], blood pressure from photoplethysmography [20], and hand-vein patient identification [21]. Vision screening is a gap in that list: the Snellen chart and the Ishihara plates need a literate, cooperative subject who reports what they see. [RF: where vision screening sits in a community health worker's day] The second thread is the wearable as a window on state, from iCalm [8] through sensor-plus-app systems for cognitive behavioural therapy [22]–[24] to the just-in-time adaptive intervention [25]. An eye-strain signal from the headband doing the imaging is a candidate trigger for the simplest such intervention: rest your eyes. [RF: JITAI framing]

III. System

Hardware. Muse 2 at 256 Hz over Bluetooth to a phone, forwarded to the laptop over UDP; a gold cup with paste at Oz on the Muse auxiliary input, held by the headset strap; a Quest 3S; a laptop. Under $600, nothing modified. [TODO: photograph of the set-up]

Stimulus. A web page served by the laptop and opened in the headset's browser. One square flickers between black and white, or black and a colour, at the target position; a dot marks it and the subject follows the dot. The page flips on the display's frame clock, so at 72 fps the 12 Hz flicker is exactly three frames on and three off, and 9 Hz four and four; it measures and reports the delivered frequency, and reports its visibility, since a background tab renders at one frame per second.

Session. An eyes-closed alpha check confirms the cup is on the scalp. Calibration is six blocks of 8 s full-field flicker on and off; a line statistic at the flicker frequency with an exact permutation test, and a d' per electrode, decide whether to proceed and how to weight the electrodes. The scan visits every grid position in serpentine order, 8 s each, with rest breaks on the subject's trigger, then revisits the positions whose black-or-white decision is uncertain. Each visit's pixel is the paper's whole-dwell ratio, power at the flicker frequency and its harmonic over the 14 to 50 Hz floor, weighted across electrodes; visits are averaged and the picture thresholded by Otsu's method in the log domain [26]. Every picture is scored against the target by correlation and against nulls in which the EEG is shifted in time by tens of seconds, so the same analysis runs on data that cannot contain the image.

Colour by planes. Red, green and blue are three scans, each black-versus-colour with its own calibration; the three threshold decisions per cell give one of eight colours. Blue is weak (about 0.4 of red in power at the same contrast), so it gets a frequency sweep before the scan (6, 7.2, 9, 12 Hz; 9 Hz won on every sitting), a patch two cells wide, and a second visit to its ten least certain cells. Green gets a second pass; red, which has never missed a cell, one.

Guards. Each of these cost a session before it was caught: a laptop on battery entering standby (the driver now holds the machine awake and redoes a position after a gap); the driver's own 20 s of scoring reported as a freeze (it checks its CPU time first); a browser tab restored from the previous session in front of the new one (a session token and a new-task launch); an orphaned recorder holding the UDP port; an ear electrode coming loose at minute 11 (a loose-channel guard is next). A synthetic subject that emits the right SSVEP for the stimulus log exercises the whole stack without a person.

IV. Results

All results are from one subject, the first author, September to October 2026.

Where the eye answers. The first sweep (Fig. 1) put the response between 7.5 and 12 Hz: Oz line ratios of 9.0 at 10.4 Hz and 5.9 at 11.4 Hz against 1.0 with the screen off, nothing at 20 Hz. Mann's 15 Hz sits on the weak side of this curve, so our sessions use 12 Hz, the nearest frequency the display delivers exactly. A stock Muse without the cup showed no response. Measure the subject first.

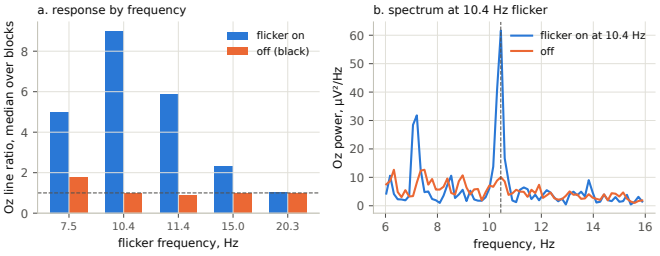


Fig. 1. The subject's response by frequency on the laptop (left) and the Oz spectrum at 10.4 Hz flicker (right): a single-bin line with the alpha hump unchanged.

TABLE I
THE EIGHT-COLOUR CHART (8 BY 4) OVER THREE SITTINGS: CELLS RIGHT PER PLANE, CELLS IN THE RIGHT COLOUR, COLOURED CELLS WITH THE RIGHT HUE, AND COLOUR r WITH THE LARGEST TIME-SHIFTED NULL.

sitting	R	G	B	right	hue	r (null)
2026-10-10 17:57	32	28	28	24/32	19/24	0.79 (0.32)
2026-10-10 19:50	29	31	23	20/32	16/24	0.66 (0.32)
2026-10-12 10:40	32	31	23	23/32	16/24	0.75 (0.25)

A legible word. "NO" in a pixel font on 9 by 5 positions (Fig. 2): $r = 0.79$ after one pass, 0.88 after two, 0.91 after three, nulls at most 0.23, 43 of 45 positions right after the threshold, 18 minutes. A square 4.6 times larger reached 0.87 in two passes, so a larger square did not buy signal; a scalable font squeezed onto 48 positions was a blob before any brain was involved ($r = 0.41$ to 0.49 against nulls to 0.48).

Eight colours over three sittings. Table I and Fig. 3 and 4. The first sitting read red 32, green 28 and blue 28 of 32, giving 24 of 32 cells in the right colour and the hue right on 19 of 24 coloured cells, $r = 0.79$ against nulls of 0.32. Two hours later, 20 of 32: green's second pass helped, blue's second pass was lost to the right ear electrode coming loose at minute 11. Two days later, 23 of 32, red perfect again, blue lost on the right half of the field with every electrode holding. One wrong cell in four on the chart is almost entirely blue's.

A sealed acuity read-out. A tumbling E and a ring that is closed (O) or missing its bottom (U) sit on a 12 by 7 grid; one of the eight combinations is drawn at random and sealed before the scan. A template correlation on the EEG picture names the orientation and the ring, then the seal is opened. On the first human run (Fig. 5) the picture reached $r = 0.89$, 80 of 84 cells, and the read-out said E up, ring U, with margins of 0.24 and 0.22 over the runners-up; the seal agreed. One trial at a one-in-eight chance is a demonstration, not an acuity measurement; a series over optotype sizes is the next step [11].

Toward a real photograph. The 2019 portrait went in thresholded to two tones and came back a silhouette. We prepare the same at our grid and add a forced choice: five portraits (Fig. 6) cut to 16 by 16 cells at four grey levels,

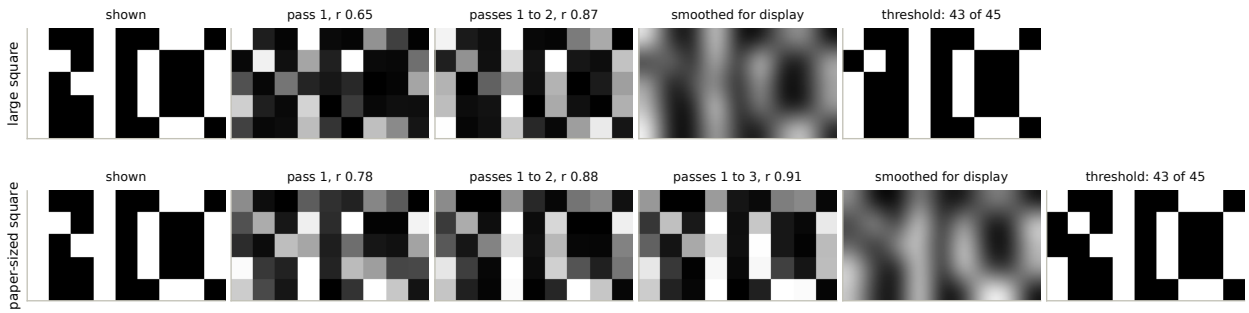


Fig. 2. "NO" read from the EEG in the Quest 3S, 12 Hz, 8 s per position, with a large square (top) and the paper-sized square (bottom): shown, after each pass, smoothed as in the 2019 paper, and thresholded. Both reach 43 of 45.

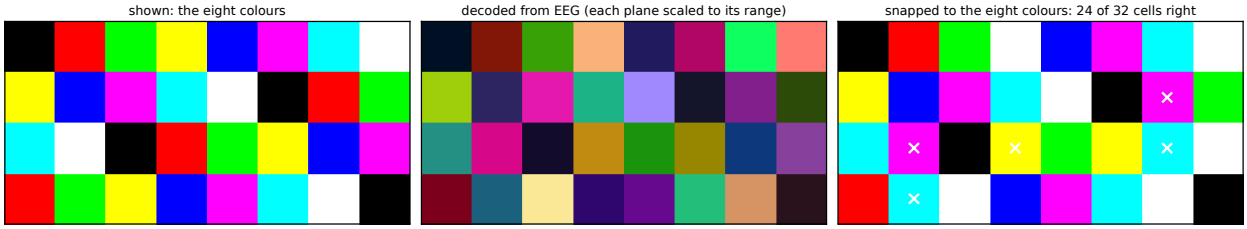


Fig. 3. Eight colours on the first colour sitting: shown, decoded with each plane scaled to its range, and snapped to the eight colours with wrong cells marked. 24 of 32.

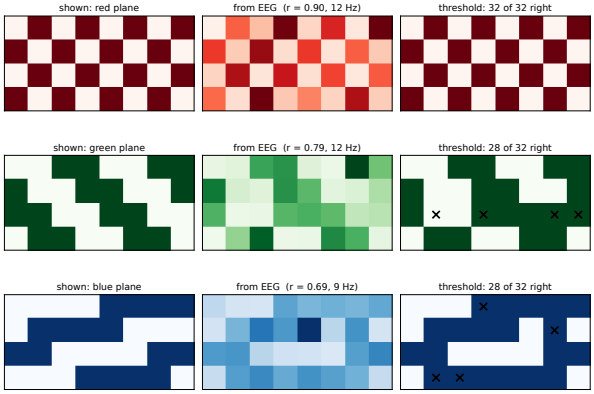


Fig. 4. The same sitting plane by plane: shown, from the EEG, thresholded. Red 32, green 28, blue 28 of 32; blue at 9 Hz with the patch doubled.

the block-portrait size at which a face is recognisable [27], [28]; one is drawn and sealed, and the read-out names the person by correlation against all five. On the clean targets the choice is right 100% of the time with a quarter of the cells flipped, chance one in five. A colour photograph does not survive the three planes (its colours sit mid-cube and flip), but a flat illustration in the eight corner colours does (Fig. 6, right). [TODO: wedge result, first sealed photograph, officer scan]

Eye strain from the same recording. After two colour sittings in one evening the subject reported irritated eyes, a slight forehead ache and no drowsiness. On a Muse under a headset the blink is best seen through the Fpz reference on the far electrodes, as a shared deflection on TP9, TP10 and the cup. Counted so, the blink rate was 6.8 per minute in the first sitting and 12.1 in the second against 3 to 4 on a reference day, with amplitude and duration unchanged. Within a sitting (Fig. 7) blinks nearly stop while a cell flickers and burst within a second of it going dark, 50 to 65 per minute in that second, a suppress-and-release pattern the reference day lacks [29], [30]. Drowsiness markers were flat: no rise in frontal theta, no long blinks; an earlier report of rising alpha was the loose ear electrode and is retracted. The forehead ache is not in the EEG (frontalis 1.5 to 2.3 μV rms); the headset's facial interface is the likelier cause. The SSVEP itself fell by half between the blue planes of the two sittings (Fig. 8), and the cup's noise rose from minute 20 of the second, so part of what reads as fatigue is contact.

V. Toward a Screening Tool for the Field

What a community health worker would do with this decides the design. Three readings come from one sitting under half an hour. A *colour response profile*: the per-colour calibrations are three numbers before any scan (red, green, blue line statistics of +5.0, +4.3, +5.5 on the first sitting), the raw material of an objective colour-vision screen once the stimuli are isoluminant, which our smooth-flicker mode can display, and there is more than one subject. [RF: what a colour

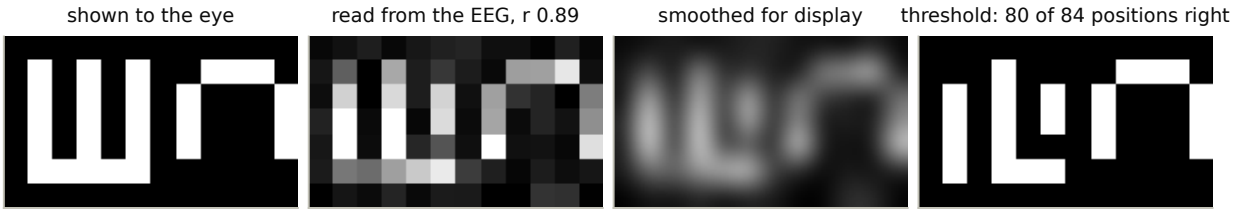


Fig. 5. The sealed pair read from the EEG: shown, one value per position, smoothed, thresholded. The read-out named E up and ring U before the seal was opened; both right.

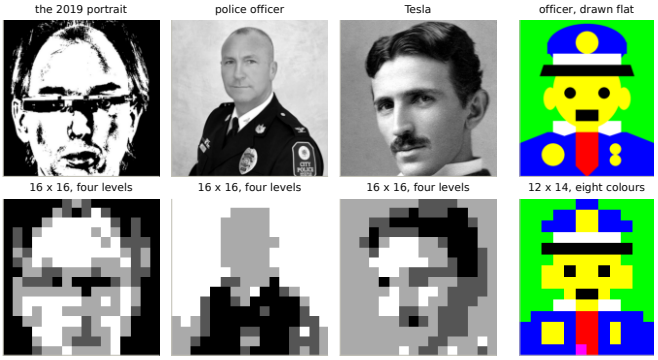


Fig. 6. The photograph gallery: the crop the eye gets over its 16 by 16 four-level target, for Mann's 2019 portrait, a police officer and Tesla (Lincoln and Einstein complete the set); right, the officer as a flat illustration and its 12 by 14 poster for the colour planes.

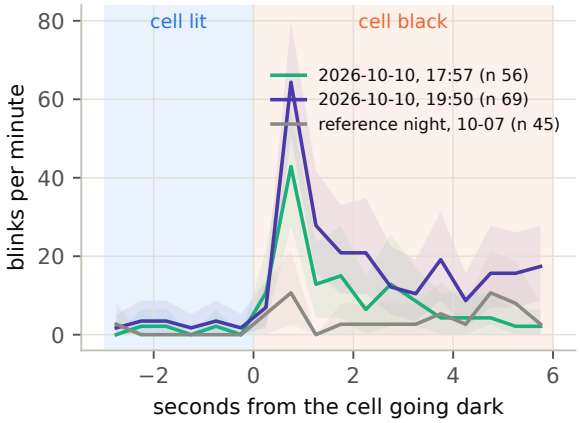


Fig. 7. Blink release when the cell goes dark: blinks per minute in half-second bins around the moment a flickering cell turns black, two colour sittings against a reference day. Bands are 95% block-bootstrap intervals.

screen must report for a worker to act on it] An *acuity read-out*: the sealed optotype, then a threshold series, giving acuity without a verbal response, as the sweep VEP does in a clinic [11]. An *eye-strain reading and a prompt*: the blink pattern and rate are live from the same headband, and a "rest your eyes" message is the simplest just-in-time intervention [25]. [RF: prompt design and what to log around it]

Limits. One subject. One cell in four wrong on a 32-cell

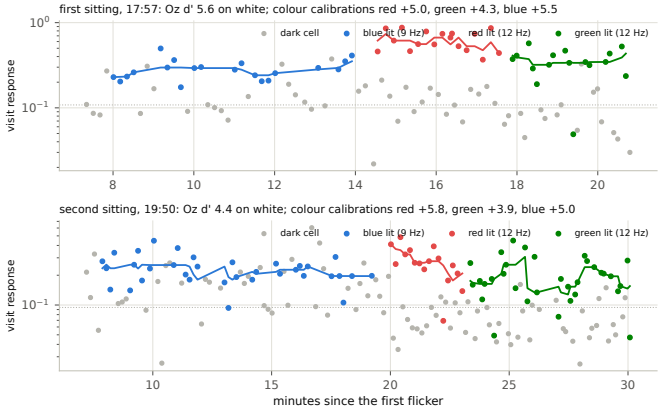


Fig. 8. The SSVEP of every lit cell through the two sittings of one evening (top the first, bottom the second), by plane. The blue response fell by half between them.

colour chart; a 96-cell picture is an hour. Blue is fragile, and 9 Hz sits next to alpha. Contact is the largest practical risk: the cup's d' ranged from 0.8 to 5.6 across days, and one loose electrode cost a plane. Flicker at 3 to 30 Hz can provoke seizures in photosensitive epilepsy and the method must not be used with anyone with that history. The eye-strain result is one evening without a control. [RF: field constraints: time per subject, lighting, who holds the laptop] Next: two more subjects, the acuity series, the loose-channel guard, the sealed photograph, the prompt.

VI. Conclusion

The eye as a camera works on a $250 headband with one added cup and a $300 headset: a legible word, eight colours over three sittings, and an optotype named from the EEG before its seal was opened, from a subject who only looked at a dot. The recording that gives the image also gives the eye strain. That is the start of a visual screener in the low-cost, objective, worker-operated tradition; the rest is subjects, stimuli that separate colour from brightness, and a prompt that closes the loop.

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[TODO: MannLab Canada; Interaxon for the auxiliary-electrode documentation. Confirm wording.]

REFERENCES

[1] S. Mann *et al.*, "The human eye as a camera," in *Proc. IEEE Int. Conf. on E-health Networking, Application and Services (HealthCom)*, 2019, wearcam.org/eyecam.pdf.

[2] ——, "Eye itself as a camera: Sensors, integrity, and trust," in *Proc. 5th ACM Workshop on Wearable Systems and Applications (WearSys)*, 2019.

[3] A. M. Norcia, L. G. Appelbaum, J. M. Ales, B. R. Cottereau, and B. Rossion, "The steady-state visual evoked potential in vision research: A review," *Journal of Vision*, vol. 15, no. 6, p. 4, 2015.

[4] F.-B. Vialatte, M. Maurice, J. Dauwels, and A. Cichocki, "Steady-state visually evoked potentials: Focus on essential paradigms and future perspectives," *Progress in Neurobiology*, vol. 90, no. 4, pp. 418–438, 2010.

[5] R. R. Fletcher and M.-Z. Poh, "Wearable sensors: Opportunities and challenges for low-cost health care," in *Proc. IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2010.

[6] R. R. Fletcher, X. Soriano, H. Bajaj, and S. Gosh-Jerath, "Development of smart-phone based child health screening tools for community health workers," in *Proc. IEEE Global Humanitarian Technology Conference (GHTC)*, 2017.

[7] A. Anand, D. Chamberlain, R. Kodgule, and R. R. Fletcher, "Pulmonary screener: A mobile phone screening tool for pulmonary and respiratory disease," in *Proc. IEEE Global Humanitarian Technology Conference (GHTC)*, 2018.

[8] R. R. Fletcher, K. Dobson, M. S. Goodwin, H. Eydgahi, O. Wilder-Smith, D. Fernholz, Y. Kuboyama, E. B. Hedman, M.-Z. Poh, and R. W. Picard, "iCalm: Wearable sensor and network architecture for wirelessly communicating and logging autonomic activity," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 215–223, 2010.

[9] R. R. Fletcher, "Engineering opportunities and challenges for the treatment of mental health and behavioral medicine," in *Proc. IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2014.

[10] A. Armengol-Urpi and S. E. Sarma, "Sublime: A hands-free virtual reality menu navigation system using a high-frequency SSVEP-based brain-computer interface," in *Proc. 24th ACM Symposium on Virtual Reality Software and Technology (VRST)*, 2018.

[11] A. M. Norcia and C. W. Tyler, "Spatial frequency sweep VEP: Visual acuity during the first year of life," *Vision Research*, vol. 25, no. 10, pp. 1399–1408, 1985.

[12] J. Rabin, E. Switkes, M. Crognale, M. E. Schneck, and A. J. Adams, "Visual evoked potentials in three-dimensional color space: Correlates of spatio-chromatic processing," *Vision Research*, vol. 34, no. 20, pp. 2657–2671, 1994.

[13] O. E. Krigolson, C. C. Williams, A. Norton, C. D. Hassall, and F. L. Colino, "Choosing MUSE: Validation of a low-cost, portable EEG system for ERP research," *Frontiers in Neuroscience*, vol. 11, p. 109, 2017.

[14] S. Mann, A. Vicol, N. Sabbour, G. Wodu, K. Monaragala, X. Chen, and P. Aniolowski, "State-of-float: From balling to future space travel," *Proc. IEEE Int. Conf. on Consumer Electronics (ICCE)*, 2026.

[15] D. Chamberlain, R. Kodgule, and R. R. Fletcher, "A mobile platform for automated screening of asthma and chronic obstructive pulmonary disease," in *Proc. IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2016.

[16] C. Infante, D. Chamberlain, Y. Thorat, R. Kodgule, and R. R. Fletcher, "Use of cough sounds for diagnosis and screening of pulmonary disease," in *Proc. IEEE Global Humanitarian Technology Conference (GHTC)*, 2017.

[17] R. R. Fletcher, N. Pignatelli, A. Jimenez-Galindo, and S. Gosh-Jerath, "Development of smart phone tools for printed diagnostics: Challenges and solutions," in *Proc. IEEE Global Humanitarian Technology Conference (GHTC)*, 2016.

[18] R. R. Fletcher, G. Schneider, L. Bikorimana, G. Rukundo, A. Niyigena, E. Miranda, R. Riviello, F. Kateera, and B. Hedt-Gauthier, "The use of mobile thermal imaging and deep learning for prediction of surgical site infection," in *Proc. IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2021.

[19] S. Byun, B. G. B. Bueno, Y. Gupta, N. Dhadge, S. Pawar, R. Kodgule, and R. R. Fletcher, "The use of thermal imaging and deep learning for pulmonary diagnostics and infection detection," in *Proc. IEEE Int. Conf. on Body Sensor Networks (BSN)*, 2021.

[20] M. Elgendi, R. R. Fletcher, Y. Liang, N. Howard, N. H. Lovell, D. Abbott, K. Lim, and R. Ward, "The use of photoplethysmography for assessing hypertension," *npj Digital Medicine*, vol. 2, no. 1, pp. 1–11, 2019.

[21] R. R. Fletcher, V. Raghavan, and R. Zha, "Development of mobile-based hand vein biometrics for global health patient identification," in *Proc. IEEE Global Humanitarian Technology Conference (GHTC)*, 2014.

[22] R. R. Fletcher, S. Tam, O. Omojola, R. Redemske, and J. Kwan, "Wearable sensor system and mobile application for use in cognitive behavioral therapy for drug addiction and PTSD," in *Proc. IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2011.

[23] E. W. Boyer, R. R. Fletcher, R. F. Fay, D. Smelson, D. Ziedonis, and R. W. Picard, "Preliminary efforts directed toward the detection of craving of illicit substances: The iHeal project," *Journal of Medical Toxicology*, 2012.

[24] N. R. Leonard, B. Casarjian, R. R. Fletcher, C. Prata, D. Sherpa, A. Kelemen, S. Rajan, R. Salaam, C. Cleland, and M. Gwadz, "Theoretically-based emotion regulation strategies using a mobile app and wearable sensor among homeless adolescent mothers: Acceptability and feasibility study," *JMIR Pediatrics and Parenting*, vol. 1, 2018.

[25] I. Nahum-Shani, S. N. Smith, B. J. Spring, L. M. Collins, K. Witkiewitz, A. Tewari, and S. A. Murphy, "Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support," *Annals of Behavioral Medicine*, vol. 52, no. 6, pp. 446–462, 2018.

[26] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[27] L. D. Harmon and B. Julesz, "Masking in visual recognition: Effects of two-dimensional filtered noise," *Science*, vol. 180, no. 4091, pp. 1194–1197, 1973.

[28] L. D. Harmon, "The recognition of faces," *Scientific American*, vol. 229, no. 5, pp. 70–82, 1973.

[29] J. A. Stern, D. Boyer, and D. Schroeder, "Blink rate: A possible measure of fatigue," *Human Factors*, vol. 36, no. 2, pp. 285–297, 1994.

[30] J. E. Sheedy, J. Hayes, and J. Engle, "Is all asthenopia the same?" *Optometry and Vision Science*, vol. 80, no. 11, pp. 732–739, 2003.